

Assignment: Subdomain Enumeration and Website Technology Analysis

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Introduction

In today's digital world, ensuring the security of websites is extremely important. Ethical hackers and cybersecurity researchers begin their work by gathering information about a target system. This process is known as reconnaissance. In this assignment, I performed subdomain enumeration and technology analysis on the website of Tesla, Inc..

Practical Demonstration (Lab-Based Analysis)

As part of this assignment, a controlled lab environment was used on Kali Linux to understand how publicly available breach data can be analyzed for educational purposes.

The tool used in this process was:

Breach-Parse

Breach-Parse is a tool designed to process large datasets from publicly available breach compilations. It helps security researchers analyze exposed credentials in an ethical and controlled environment.

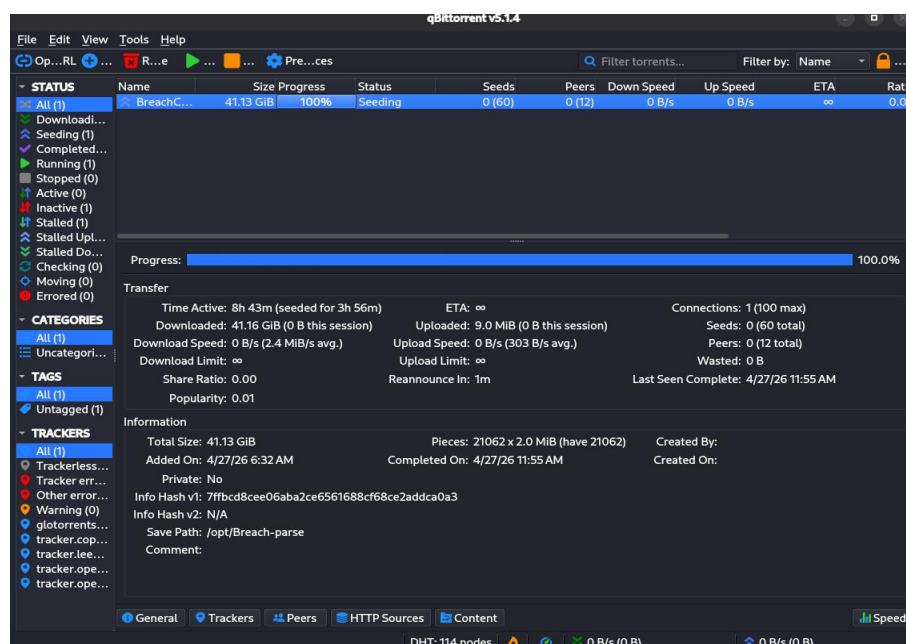


Figure 1.1

```
(kali@kali)-[~]
$ ls
192.168.226.133  cyberFight.txt.save  FootprintProject  go  main.cpp.save  Pictures  subdomains.txt
angkon          data                fork.1            hay.txt  Music          ppp        Templates
angku           ddd.txt            fork3             hi      myself.txt    program   urls.txt
'Bash Scripting' Desktop           fork3.c          iso27001_scanner  new2file.txt  program.c  Videos
CamPhish       Documents          fork4            iso_gap_tool      newfile.txt   program.c.save
Capsuleit      Downloads          fork4.c          iso-scanner       ok.txt        Public
cyberFight.txt Firefox_wallpaper.png fork5.c.save      iso_scanner      output.txt    subdomains1.txt

(kali@kali)-[~]
$ /opt

(kali@kali)-[/opt]
$ ls
Breach-parse  microsoft

(kali@kali)-[/opt]
$ cd Breach-parse

(kali@kali)-[/opt/Breach-parse]
$ ls
BreachCompilation  breach-parse.sh  tesla-master.txt  tesla-passwords.txt  tesla-users.txt
```

Figure 1.2

```
(kali@kali)-[/opt/Breach-parse]
$ cat tesla-master.txt
cxrscommon@gmail.txt1@gmail.com:86da4cc9477ec6e222f40137e9710293
cxrscommon@gmail.txt1@gmail.com:d68pyfuH2V

(kali@kali)-[/opt/Breach-parse]
$ cat tesla-passwords.txt
86da4cc9477ec6e222f40137e9710293
d68pyfuH2V

(kali@kali)-[/opt/Breach-parse]
$ cat tesla-users.txt
cxrscommon@gmail.txt1@gmail.com
cxrscommon@gmail.txt1@gmail.com

(kali@kali)-[/opt/Breach-parse]
$ ./breach-parse.sh
Breach-Parser v2: A Breached Domain Parsing Tool by Heath Adams

Usage: ./breach-parse.sh <domain to search> <file to output> [breach data location]
Example: ./breach-parse.sh @gmail.com gmail.txt
Example: ./breach-parse.sh @gmail.com gmail.txt "~/Downloads/BreachCompilation/data"
You only need to specify [breach data location] if its not in the expected location (/opt/breach-parse/BreachCompilation/data)

For multiple domains: ./breach-parse.sh "<domain to search>|<domain to search>" <file to output>
Example: ./breach-parse.sh "@gmail.com|@yahoo.com" multiple.txt
```

Figure 1.3

Explanation of Output Files

- **tesla-users.txt**
Contains a list of usernames or email addresses found in the dataset.
- **tesla-passwords.txt**
Contains extracted passwords associated with the dataset.
- **tesla-master.txt**
A combined file that may include both usernames and passwords in structured format.

What is Subdomain Enumeration?

Subdomain enumeration is the process of identifying subdomains associated with a main domain (for example: tesla.com). Subdomains may include services like mail, shop, or internal systems.

Purpose:

- To discover hidden or less visible parts of a website
- To understand the overall structure of a domain
- To expand the scope of security testing

Tools Used for Subdomain Enumeration

1. Sublist3r

Sublist3r is an automated tool used to find subdomains using search engines and other online sources.

Features:

- Fast scanning
- Passive information gathering
- Easy to use

```
(kali@kali)-[~]
└─$ sudo su
[sudo] password for kali:
(kali@kali)-[/home/kali]
└─# apt install sublist3r
sublist3r is already the newest version (1.1-5).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 438
(kali@kali)-[/home/kali]
└─# sublist3r -d openai.com

          SUBLIST3R
          # Coded By Ahmed Aboul-Ela - @aboul3la

[-] Enumerating subdomains now for openai.com
[-] Searching now in Baidu..
[-] Searching now in Yahoo..
[-] Searching now in Google..
```

Figure 2.1

```

[-] Searching now in Yahoo..
[-] Searching now in Google..
[-] Searching now in Bing..
[-] Searching now in Ask..
[-] Searching now in Netcraft..
[-] Searching now in DNSDumpster..
[-] Searching now in Virustotal..
[-] Searching now in ThreatCrowd..
[-] Searching now in SSL Certificates..
[-] Searching now in PassiveDNS..
[-] DNSDumpster module failed: Could not find CSRF token on DNSDumpster page
[-] Error: Virustotal probably now is blocking our requests
[-] Total Unique Subdomains Found: 42
www.openai.com
academy.openai.com
api.openai.com
chat.gateway.unified-87.api.openai.com
auth.openai.com
external.auth.openai.com
setup.auth.openai.com
auth0.openai.com
beta.openai.com
cdn.openai.com
chat.openai.com
community.openai.com
cookbook.openai.com
deploymentsafety.openai.com
developers.openai.com
em.openai.com
up13243.email.openai.com
evals.openai.com
feather.openai.com
xqoms56vz6t42xp.c.feather.openai.com

```

Figure 2.2

Subfinder

Subfinder is a powerful passive subdomain discovery tool widely used in cybersecurity.

Features:

- Uses multiple data sources
- Provides accurate results
- Popular in bug bounty hunting

```

(root@kali)-[/home/kali]
# subfinder -h
Subfinder is a subdomain discovery tool that discovers subdomains for websites by using passive online sources

Usage:
  subfinder [flags]

Flags:
INPUT:
  -d, -domain string[]  domains to find subdomains for
  -dL, -list string      file containing list of domains for subdomain discovery

SOURCE:
  -s, -sources string[]  specific sources to use for discovery (-s crtsh,github). Use -ls to display
  -recursive              use only sources that can handle subdomains recursively rather than both re
  -all                    use all sources for enumeration (slow)
  -es, -exclude-sources string[] sources to exclude from enumeration (-es alienvault,zoomyeapi)

FILTER:
  -m, -match string[]    subdomain or list of subdomain to match (file or comma separated)
  -f, -filter string[]    subdomain or list of subdomain to filter (file or comma separated)

RATE-LIMIT:

```

Figure 3.1

```
(root@kali)-[/home/kali]
# subfinder -d https://chat.openai.com/

projectdiscovery.io

[INF] Current subfinder version v2.13.0 (outdated)
[INF] Loading provider config from /root/.config/subfinder/provider-config.yaml
[INF] Enumerating subdomains for chat.openai.com
android.chat.openai.com
new.chat.openai.com
ios.chat.openai.com
auth0.chat.openai.com
chat.openai.org.chat.openai.com
staging.chat.openai.com
tcr9i.chat.openai.com
snnf76unzbe7.chat.openai.com
[INF] Found 8 subdomains for chat.openai.com in 1 second 626 milliseconds
```

Figure 3.2

crt.sh

crt.sh is an online service that allows users to search SSL/TLS certificate logs to find subdomains.

Features:

- Finds subdomains from certificate data
- Provides historical records
- Useful for passive reconnaissance

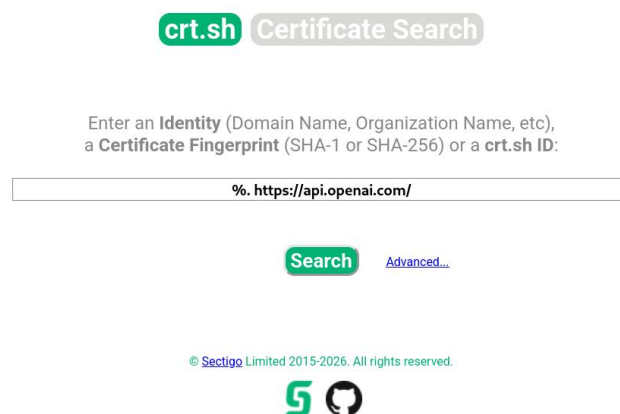


Figure 4.1

crt.sh Identity Search						
Criteria Type: Identity Match: ILIKE Search: 'openai.org' Group by Issuer						
Sorry, your search results have been truncated. It is not currently possible to sort and paginate large result sets efficiently, so only a random subset is shown below. Please retry your search with expired certificates excluded.						
Certificates	crt.sh ID	Logged At	Not Before	Not After	Common Name	Matching Identities
	7593273607	2022-09-22	2022-09-22	2022-12-21	prod-engine-aks-centralus-api-loan.api.openai.org	prod-engine-aks-centralus-api-loan.api.openai.org
	7592857006	2022-09-22	2022-09-22	2022-12-21	retool.staging.internal.api.openai.org	retool.staging.internal.api.openai.org
	7592852301	2022-09-22	2022-09-22	2022-12-21	meetings.staging.internal.api.openai.org	meetings.staging.internal.api.openai.org
	7591637748	2022-09-21	2022-09-21	2022-12-20	asr-uploader.staging.internal.api.openai.org	asr-uploader.staging.internal.api.openai.org
	7597671988	2022-09-21	2022-09-21	2022-12-20	staging-admin-westus2-output.staging.internal.api.openai.org	staging-admin-westus2-output.staging.internal.api.openai.org
	7591589656	2022-09-21	2022-09-21	2022-12-20	staging-admin-westus2-output.staging.internal.api.openai.org	staging-admin-westus2-output.staging.internal.api.openai.org
	7597664437	2022-09-21	2022-09-21	2022-12-20	humio.admin.staging.internal.api.openai.org	humio.admin.staging.internal.api.openai.org
	7591586281	2022-09-21	2022-09-21	2022-12-20	humio.admin.staging.internal.api.openai.org	humio.admin.staging.internal.api.openai.org
	7597585441	2022-09-21	2022-09-21	2022-12-20	beta42.api.openai.org	beta42.api.openai.org
	7591542569	2022-09-21	2022-09-21	2022-12-20	beta42.api.openai.org	beta42.api.openai.org

Figure 4.2

Website Technology Analysis

Wappalyzer

Wappalyzer is a browser extension and tool used to identify the technologies used in a website.

It can detect:

- Frontend frameworks (e.g., React, Angular)
- Backend technologies (e.g., Node.js, PHP)
- Content Management Systems (CMS)
- Analytics and tracking tools

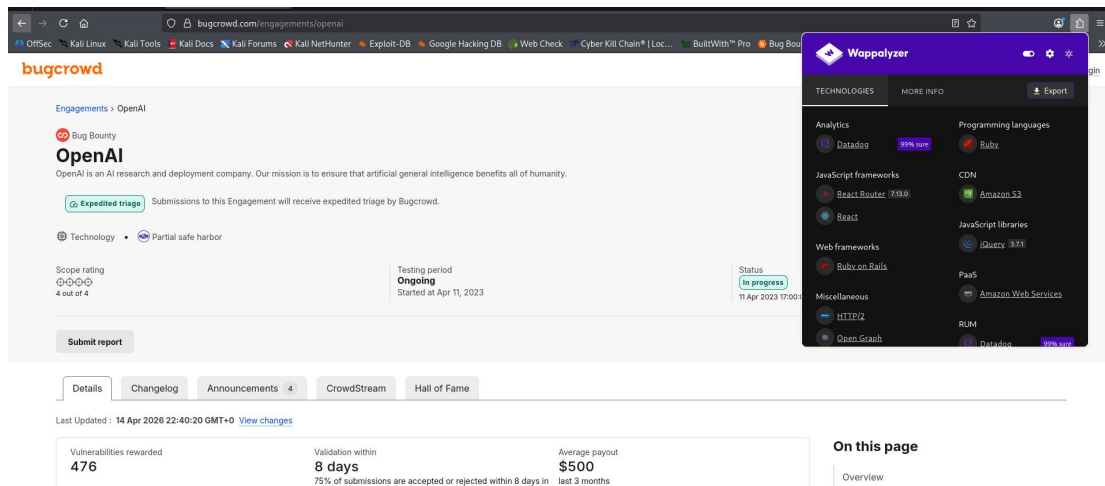


Figure 5.1

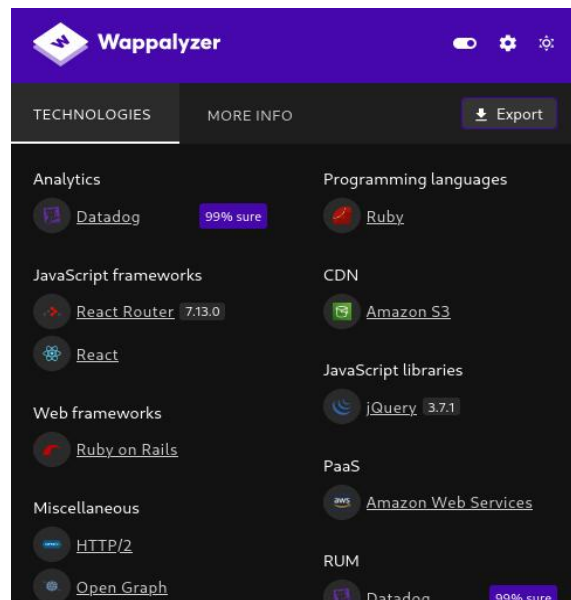


Figure 5.2

Methodology

The following steps were followed during this assignment:

- Selected the target domain (tesla.com)
- Used Sublist3r to gather initial subdomains
- Used Subfinder to collect additional subdomains
- Queried crt.sh for more subdomain data from SSL certificates
- Analyzed the website using Wappalyzer
- Collected and organized the results for analysis

Importance of This Process

- Helps identify potential security vulnerabilities
- Useful in bug bounty programs
- Provides a better understanding of website infrastructure
- Assists in ethical hacking and penetration testing

Conclusion

This assignment demonstrates the process of subdomain enumeration and website technology analysis. These techniques are essential in the field of Cybersecurity and play a crucial role in identifying and securing web applications. By using tools like Sublist3r, Subfinder, crt.sh, and Wappalyzer, researchers can effectively gather valuable information about a target system.